Medicare Part B

Fraud, Waste & Abuse Detection System

Work Sample | Portfolio Project

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Industrial Engineering | Data Engineering Certificate

1.27M	$13.95B	150K	6	51	3
Part B Records	Medicare Payments	Providers Scored	States	Risk Features	Model Layers

1. The Problem

Medicare processes over **1 billion claims annually**. Within that volume, Fraud, Waste, and Abuse (FWA) costs the federal government an estimated **$60-90 billion per year**. CMS and OIG maintain detection programs, but the scale of the data demands automated screening tools that triage providers before human investigators are engaged.

This project builds an end-to-end FWA detection system using real public CMS data, domain-specific features grounded in **industrial engineering principles**, a three-layer ML ensemble, and an investigator dashboard. Every component is built to production standards: data-quality gates, versioned transforms, and a 51-test pytest suite.

2. Industrial Engineering Principles Applied to Healthcare Fraud

Most FWA tools apply generic anomaly detection. This project applies IE principles that give each feature operational meaning a domain expert can defend:

IE Principle	Feature	What It Detects
Statistical Process Control (SPC)	Billing volume z-scores vs. specialty peer group	Providers statistically out-of-control vs. peers in same specialty

IE Principle	Feature	What It Detects
Peer Benchmarking (Jensen-Shannon Div.)	Procedure-mix distribution vs. specialty median	Unusual concentration in high-reimbursement HCPCS codes
Queueing Theory / Throughput Analysis	Services-per-day vs. theoretical capacity limit	Physically impossible billing rates for a single provider
Geographic Entropy	Patient zip-code dispersion index	Patient brokering — recruiting from abnormally wide service areas
Mahalanobis Distance	Multivariate outlier within specialty peer group	Anomalous combinations of billing dimensions not caught by single features

3. Data Sources

All data is **publicly available and free** — no PHI, no PII, no credentials required. Three federal sources are joined on NPI (National Provider Identifier):

Source	Contents	Scale
CMS Part B Provider and Service (data.cms.gov)	Billing volumes, avg charges, avg Medicare payments per provider per HCPCS code. Years: 2021, 2022, 2023.	1.27M service lines 150K unique providers 4,542 HCPCS codes 102 specialties 6 states
OIG LEIE Exclusions (oig.hhs.gov)	Providers excluded from Medicare and Medicaid participation. Used as noisy fraud labels for supervised model training.	83,001 exclusion records 29 matched to the 6-state population
NPPES NPI Registry (download.cms.gov)	Provider specialty taxonomy, practice location, and entity type. Enriches provider identity fields.	15,000 records sampled for 6-state coverage

Label caveat: LEIE exclusions cover license revocations, controlled-substance offenses, and other non-billing misconduct in addition to Medicare billing fraud. The supervised layer learns from this noisy signal; the unsupervised Isolation Forest catches anomalies regardless of labels.

4. System Architecture

Layer	Technology	Role
Ingestion	Python + Great Expectations	Download CMS API, LEIE, NPPES to DuckDB raw schema with automated data-quality gates
Transformation	dbt-duckdb (8 models)	staging to intermediate to marts; version-controlled SQL with built-in data tests and lineage
Feature Eng.	Python / pandas (51 features)	SPC z-scores, Jensen-Shannon divergence, throughput flags, Mahalanobis distance, geographic entropy, upcoding ratios
Model Layer 1	Isolation Forest (300 trees, 5%)	Unsupervised anomaly detection; catches novel patterns without label dependency
Model Layer 2	XGBoost scale_pos_weight 1:5,187	Supervised classifier trained on LEIE exclusion labels; handles extreme class imbalance without synthetic oversampling
Model Layer 3	NetworkX bipartite graph	Provider-HCPCS network; PageRank, degree anomaly, and dominant-procedure community detection
Ensemble	Weighted blend (25 / 50 / 25)	0-100 composite risk score; percentile-based triage tiers: Low / Moderate / Elevated / High / Critical
Explainability	SHAP TreeExplainer	Per-provider feature contributions; plain-English finding narratives displayed on the investigator dashboard
Serving	FastAPI + Streamlit	REST scoring endpoint with OpenAPI docs; investigator dashboard with ROI metrics, filters, and drill-down

Layer	Technology	Role
Quality / CI	pytest (51 tests) GitHub Actions	Unit and integration tests, ruff lint; 100% pass rate on CI

5. Investigator Dashboard

The Streamlit dashboard is the primary interface for analysts reviewing flagged providers. It surfaces risk tiers, model layer contribution breakdowns, SHAP-based finding narratives, and investigator ROI metrics on the Review Log tab.

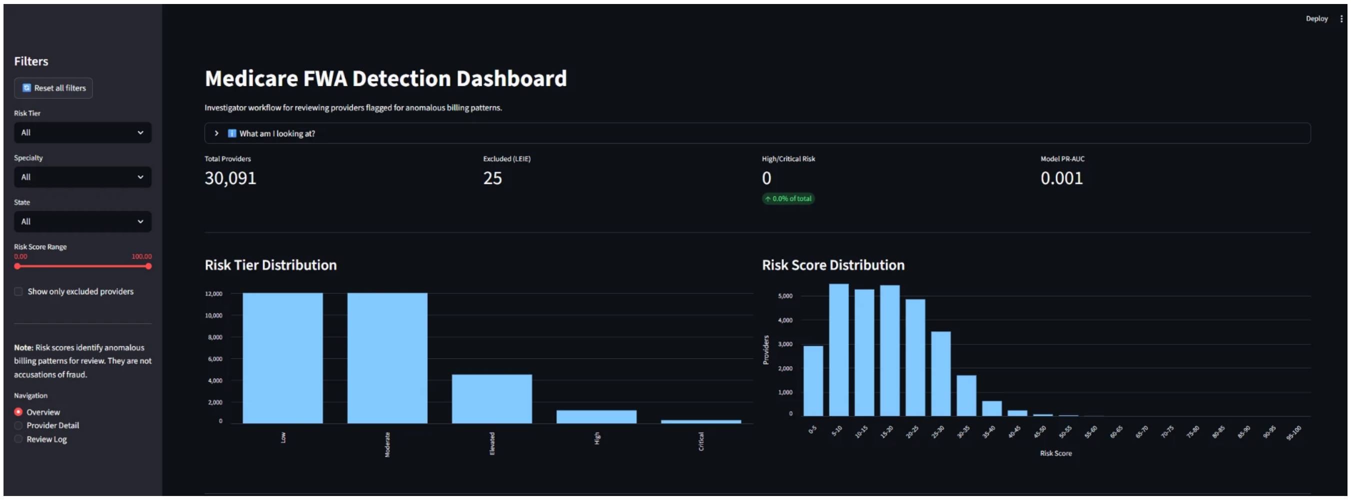
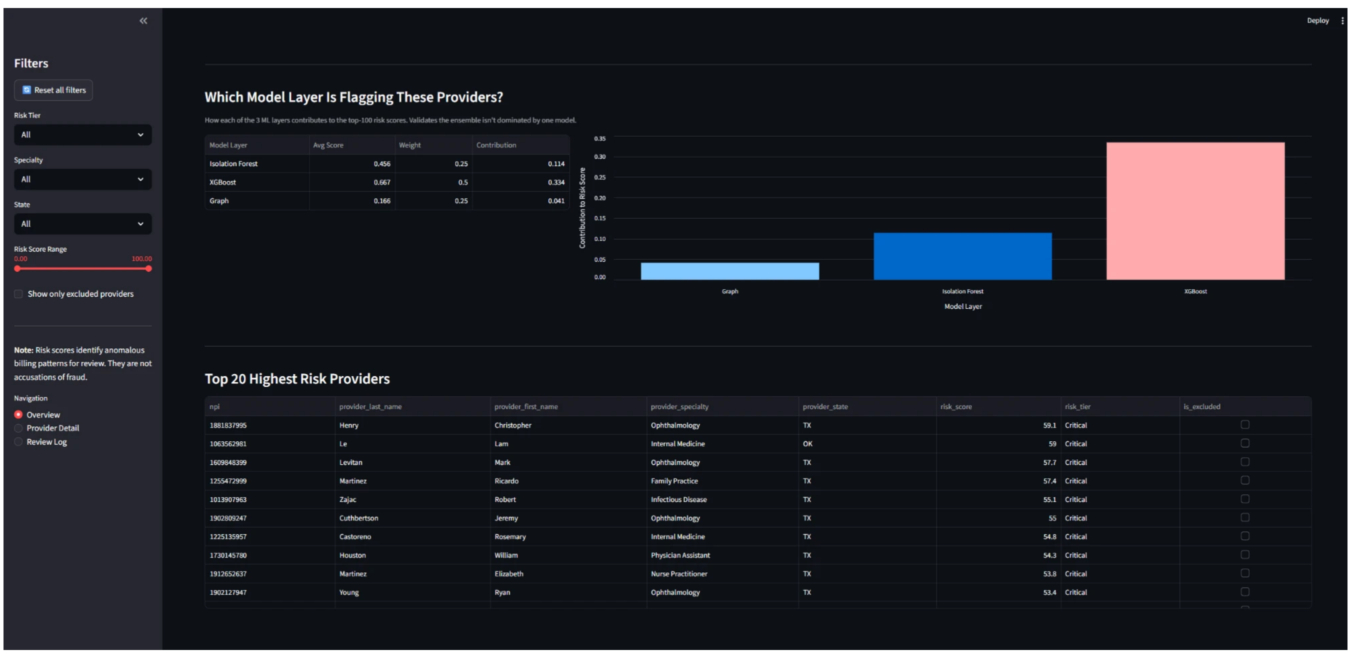


Fig 1. Overview page: 30K+ providers scored across 5 percentile-based risk tiers (Low to Critical), risk score histogram, and headline metrics.



Model Layer	Avg Score	Weight	Contribution
Isolation Forest	0.456	0.25	0.114
XGBoost	0.667	0.5	0.334
Graph	0.166	0.25	0.041

npi	provider_last_name	provider_first_name	provider_specialty	provider_state	risk_score	risk_tier	is_excluded
1881837995	Henry	Christopher	Ophthalmology	TX	59.1	Critical	
1063562981	Le	Lam	Internal Medicine	OK	59	Critical	
1609848399	Levitan	Mark	Ophthalmology	TX	57.7	Critical	
1255472999	Martinez	Ricardo	Family Practice	TX	57.4	Critical	
1013907963	Zajac	Robert	Infectious Disease	TX	55.1	Critical	
1902809247	Cuthbertson	Jeremy	Ophthalmology	TX	55	Critical	
1225135957	Castoreno	Rosemary	Internal Medicine	TX	54.8	Critical	
1730145780	Houston	William	Physician Assistant	TX	54.3	Critical	
1912652637	Martinez	Elizabeth	Nurse Practitioner	TX	53.8	Critical	
1902127947	Young	Ryan	Ophthalmology	TX	53.4	Critical	

Fig 2. Model layer contribution breakdown (Isolation Forest, XGBoost, Graph) and Top-20 highest-risk providers ranked by composite score.

6. Key Results

The system functions as a **peer-relative triage tool**. The top-25 risk-scored providers show substantially elevated billing activity versus the overall population:

Metric	Top-25 Flagged Providers	vs. Population
Median Medicare payments	$1.02M	40.7x population median
Median services rendered	4,966	11.9x population median
Median services / beneficiary	5.0	3.6x population median
LEIE class imbalance	--	1 : 5,187
Model PR-AUC	0.001	Baseline: 1/5,187 = 0.0002

Honest framing: PR-AUC is intentionally low. With only 29 LEIE-matched providers out of 150K, meaningful supervised lift is limited by label quality, not model design. The system's practical value is surfacing extreme statistical outliers for investigator review — the same function CMS fraud-analytics teams perform in practice. A production deployment would use CMS payment-suspension data and DOJ strike-force enforcement labels, which are not publicly available.

7. Technical Stack

Component	Choice	Decision Rationale
Storage	DuckDB	Columnar analytics on laptop; SQL dialect maps directly to Postgres/Snowflake in production
Transforms	dbt-duckdb	Version-controlled SQL with built-in testing and data lineage; analytics engineering standard
ML	scikit-learn + XGBoost	Production-grade tabular ML; widely understood in government analytics contexts
Graph	NetworkX	Bipartite provider-HCPCS network; PageRank and community features without GPU requirement
Explainability	SHAP TreeExplainer	Required for government/regulated use; investigators need to understand why a provider was flagged
API	FastAPI	Async, auto-generated OpenAPI docs, Pydantic validation; easily containerised with Docker
Dashboard	Streamlit	Rapid investigator-workflow prototyping; interactive filters, drill-down, and review logging
Quality	pytest + ruff + Great Exp.	51 unit tests, lint CI via GitHub Actions, data contracts on every ingestion run

8. What I Would Do Differently in Production

- **Multi-year temporal splits:** Train on 2020-2021, validate on 2022, deploy for 2023. Monitor for concept drift quarterly.
- **Better labels:** Replace LEIE with CMS payment suspensions and DOJ healthcare-fraud strike-force takedowns — billing-fraud-specific labels rather than all exclusion types.
- **Claims-level features:** Access to individual claims via CMS DUA would enable temporal billing-pattern analysis, beneficiary-level features, and diagnosis-procedure consistency checks.
- **Active learning loop:** The dashboard Review Log captures investigator decisions. These should feed back into model retraining as investigators confirm or dismiss flags.
- **Role-based access and audit logging:** Investigators should only see providers in their assigned region and specialty, with every score lookup logged for compliance.